Supplementary Material
An Inherent Blind Spot of Physics-Based Attestation for Irrigation Logs, and Device-Anchored Repair in Monitoring, Reporting and Verification for Low-Emission Rice

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S1. Fraud classes and what each defence observes

Table S1. The three forgeable quantities of Eq. (7) in the main text and which fraud class alters each. "true" means the quantity is preserved, so the corresponding defence is blind by Prop. 1 in the main text.

Fraud variant	Total mm	Each depth	Timing	What it forges
Honest farmer (field conditions)	true	true	true	Reference: seven reality-gap axes active
Added irrigation events	FALSE	true	true	Inflates the record of management activity
Infeasible claimed depth	FALSE	FALSE	true	One third of entries set to the whole storage span
Omission of real commands	FALSE	true	true	Deletes a 96 h window to manufacture a dry phase
Claim consolidation, 24 h	true	varies	FALSE	Same water, fewer and larger events
Claim consolidation, 96 h	true	varies	FALSE	Same water, events snapped to a coarse grid
Time shifting (rational)	true	true	FALSE	Hill-climbs each hour to maximise creditable dry phases
Rational adversary	true	true	FALSE	Chooses the consolidation window that maximises dry phases

S2. The admissible depth bound across Θ

Table S2. The admissible depth bound $u_{\max}$ (Eq. (16) in the main text) across the FAO-56 Table 2 range. The original single value 61 mm is one row of this table.

Soil (FAO-56 Table 2)	Span (mm)	$u_{\max}$ at $w_0 = 0$	$u_{\max}$ at $w_0 = 0.90$
sandy loam	55.0	49.0	54.5
loam	75.0	65.0	72.5
silt loam	90.0	77.0	86.0
silty clay loam	70.0	61.0	68.0
clay	80.0	69.0	77.0
Range to be published	—	**49.0**	**86.0**

S3. Reachability of the wilting-point filter

Table S3. Reachability of the wilting-point filter over 150 configurations with an empty log, i.e. the driest forcing available.

Station	Year	Max clipped depletion	V_2 fires	Interpretation
can tho	2024	0.998	0	never below wilting point
can tho	2025	0.988	0	never below wilting point
soc trang	2024	0.999	0	never below wilting point
soc trang	2025	0.992	0	never below wilting point
ca mau	2024	0.991	0	never below wilting point
ca mau	2025	0.980	0	never below wilting point

S4. Subsidence Invariance

AWD compliance is usually stated as an absolute water-table or ponding depth, for example 15 cm below the surface, the reflooding threshold of the Safe-AWD treatment in [1]. In a subsiding delta the surface elevation $z_s(t)$ falls at rate $\dot{z}$, so an absolute threshold measured against a fixed benchmark drifts.

Proposition 1 (Drift of an absolute threshold)**.** *Over a crediting period of T_c years an absolute threshold expressed against a fixed datum shifts by $\dot{z}\,T_c$. For $\dot{z} \in [1.1, 2.5]$ cm/yr and $T_c = 5$ yr the shift is* 5.5–12.5 *cm, that is* 37–83% *of a* 15 *cm threshold. A criterion written in the depletion fraction* (3) *has zero drift,*

because FC and WP are soil properties referenced to the root zone and not to an absolute elevation.

Proof. The absolute criterion compares a measured depth to a constant benchmark; if the benchmark and the surface move apart at $\dot{z}$, the effective threshold changes by $\dot{z}T_c$ over T_c years. The depletion criterion compares $FC - w_t$ to $(FC - WP)$, both of which are defined within the root zone and translate rigidly with the surface, so their ratio is invariant to z_s. □

The subsidence rates are those measured for the delta by [2]. The practical consequence is that a scheme anchored on depletion fraction needs no re-surveying of field benchmarks during a five-year crediting period, whereas an absolute-depth scheme does.

S5. Cost Model and Break-Even

Let a cooperative serve A hectares with N cropping seasons of device life, and let the MRV cost per hectare per season decompose as

$$C = \underbrace{\frac{n_{\text{dev}}\, p_{\text{dev}}}{A\,N}}_{\text{hardware}} + \underbrace{c_{\text{conn}}}_{\text{connectivity}} + \underbrace{h\,w}_{\text{labour}} + \underbrace{\frac{c_{\text{audit}}}{A}}_{\text{verification}}, \tag{1}$$

where n_{dev} and p_{dev} are the number and unit price of devices, h is the labour hours per hectare per season and w the rural wage. Revenue per hectare is

$$R = \kappa\,\rho, \tag{2}$$

with κ credits issued per hectare and ρ the price per credit. The scheme is economically viable when

$$C \;\le\; \beta R, \qquad \beta \in (0,1), \tag{3}$$

and $\beta = 0.5$ is the level at which monitoring consumes half of the credit revenue. Substituting (2) gives the break-even issuance

$$\kappa_{\min} = \frac{C}{\beta\rho}. \tag{4}$$

The tiers differ in n_{dev} and, more importantly, in which of the three forgeable quantities of (7) they can observe. Tier 0 is the physics layer: $n_{\text{dev}} = 0$ beyond the controller, the log is written on a handset the farmer already owns, and the filters of Sec. 4 run on public weather data at no marginal cost. Tier 1 adds one flow meter at the cooperative pumping station — $n_{\text{dev}} = 1$ for the whole area A rather than one per plot — which observes the total volume. Tier 2 adds one water-level node per cooperative, re-anchoring the twin and restoring the calibration property a measurement-based MRV needs anyway.

None of the three observes timestamps, and by Prop. 1 the first two are exactly invariant under the transformation that pays. The device anchor of Sec. 8 is therefore not an optional fourth tier but a precondition of admissibility: it is produced by the same controller that actuates the pump, so it adds storage and transmission of a record that already exists, and no additional n_{dev}.

Table S4 compares the scheme with the three architectures it replaces. With $\kappa \in [4,6]$ and $\rho \in [\$15,\$25]$ [3, 4], revenue is $R \in [\$60,\$150]$ per hectare per season, so $\kappa_{\min}$ at $\beta = 0.5$

corresponds to a monitoring budget of $30–75 per hectare. The full stack costs 8.4 per hectare per season against that 30–75 budget, i.e. 3.6 to 8.9 times below the ceiling; the margin is therefore stated as a range, not as an order of magnitude. It is also not the cheapest row of Table S4: remote sensing with field verification costs 8.0, or 0.4 less. The two answer different questions — the cheaper route confirms that a dry phase occurred but holds no per-command record against which a forged log could be reconciled — and the incremental cost of the anchor over an already-installed controller is about 1% of conservative revenue. Table S4 decomposes the comparison against the three architectures it replaces; the connectivity term is shown separately because it is the one that distinguishes a sensor-per-plot design from a single shared gateway.

Remark 1. *The device prices in Table S4 are planning estimates, not quotations. The ordering of the tiers and the break-even inequality* (3) *do not depend on them; only the absolute margin does. A quotation should replace the estimates before any procurement decision.*

Table S4. MRV cost per hectare per season for the four architectures. Monitoring dominates MRV cost, so the architecture that avoids dedicated sensing wins on labour rather than hardware.

Architecture	Hardware	Connectivity	Labour	Audit	Total $/ha	Note
Manual observation tube plus photographs	0.5	0.0	8.0	4.0	**12.5**	current practice under Decision 4801
Water-level sensor per plot	11.3	2.0	1.5	4.0	**18.8**	accurate, hard to scale to smallholders
Remote sensing plus field verification	0.0	0.0	2.0	6.0	**8.0**	revisit 2 to 4 days
Attested log, this work	3.6	1.5	0.3	3.0	**8.4**	gateway timestamp is what makes it admissible

S6. Experimental configuration

S7. Two implementation details of the robust intersection

Two implementation details of V_{rob} matter, and getting either wrong silently breaks Proposition 3 of the main text. A world that produces *no* violation must contribute the *empty set* to the intersection; skipping non-firing worlds computes $\bigcap$ over firing worlds only, which is an existential quantifier wearing the clothes of a universal one. And $\mathcal{W}$ must span the full range in which a claim can be legitimate: with $\mathcal{W} = \{0, 0.25\}$ the largest admissible depth anywhere in Θ is 22.5 mm, so every claim above that is "guilty in all worlds" and the intersection becomes vacuous — it fires on 102 of 150 worlds while 48 innocent worlds exist. Extending $\mathcal{W}$ to $\{0, 0.25, 0.5, 0.75, 0.9\}$ restores the intended semantics.

Table S5. Experimental configuration. Every value is reproducible from the released code and the public ERA5 forcing.

Component	Value
Weather	Hourly ERA5 via the Open-Meteo archive, no missing values
Stations	3: ca mau, can tho, soc trang
Climate years	2024, 2025 (2024 water-scarce, 2025 water-surplus)
Window	1 March to 29 May (2 160 h), the delta dry season
Soil textures sampled	5 across the FAO-56 Table 2 range
Reality-gap axes	7: soil, K_c stage, η, field rain, seepage, recording behaviour, w_0
Irrigation efficiency η	0.60–0.85 (water reaching roots / water pumped)
K_c regimes	fixed 1.05 (as published) and staged 1.05/1.20/0.90
Recording noise	forget 5%, round to 5 mm, one-day delay 35%, depth error 15% within $\pm$35%
Fraud variants	7 (three naive, four adversarial)
Uncertainty set $\lvert\Theta\rvert$	30 = 5 soils × 2 K_c regimes × 3 efficiencies
Initial states $\lvert w_0\rvert$	5, spanning depletion 0 to 0.90
Replays per log	24 (reduced Θ) and 150 (full Θ)
Logs audited	281

S8. Relation to prior monitoring work (full text)

AWD monitoring has been approached from three directions. *Instrumented*: a low-cost wireless water-level sensor for lowland rice under AWD [5], which solves the measurement problem but multiplies device cost per plot. *Remote sensing*: identification of AWD adoption in the Vietnamese Mekong Delta [6], water-status monitoring with ALOS-2 scattering-power decomposition [7], and region-scale machine-learning mapping of water-saving practices [8]; these cover large areas cheaply but revisit every two to four days and cannot resolve the 72 h dry phase that determines credit eligibility. *Programme level*: the one-million-hectare high-quality low-emission rice programme [9], which had brought over 350 000 hectares under cultivation by 2026 [10], and MRV identified as the missing link for low-emission rice in Southeast Asia [11].

On the agronomic side, meta-analysis quantifies the water and yield effects of AWD [12], while adoption economics [13] and the technical and institutional barriers to scaling [14] explain why instrumented monitoring has not spread. None of these lines evaluates the *reliability of the action record itself*, which is the question asked here.

The gateway supplying the device-anchored timestamp is the controller of the companion work [15], whose command dispatch runs over a burst-noise channel of the Gilbert–Elliott type [16, 17]. Our own prior scheduling results for safety-critical agricultural IoT networks address the same transmission regime [18, 19], but neither establishes an evidence record, which is the function relied on here. Because that controller already signs every command it issues, the anchor adds

storage and transmission of a record that exists regardless — which is what keeps the marginal cost of attestation near zero.

S9. Extended discussion

S9.1 Why publishing the threshold is not the problem

One might conclude that ε should be kept secret. It should not, for two reasons. A verifier must be able to reproduce a decision, which requires published parameters. And secrecy would not help: by Proposition 1 of the main text the time-shifting attack preserves the total volume and every individual depth, so it does not need to know ε in order to stay beneath the infiltration bound — it needs only to avoid moving water, which is visible to any farmer. The vulnerability is structural, not a consequence of disclosure.

S9.2 Relation to remote sensing (full text)

Satellite radar is the obvious low-cost alternative, so rather than argue about it we measured it. Table 5 of the main text scores every route by one quantity: how many of the 150 credit-eligible dry phases actually present across the 60 simulated fields it could confirm. A phase counts as confirmable only if at least one observation falls inside it, because a 72 h phase lying wholly between two passes cannot be evidenced at all — which costs the farmer a credit even when the practice was performed correctly. Single-satellite Sentinel-1, revisiting every twelve days [20], confirms 90/150 (60%); the full A+B constellation at six days [21] reaches 124/150 (83%), and still misses one phase in six at zero device cost. A continuous water-level tube confirms all 150, at one device per plot [5]. The signed command record also confirms all 150 — because the phases are defined by the intervals between actuations, which the controller already timestamps — at no additional device, since the controller is the one that opened the pump. Two cautions belong with that comparison. The confirmation column measures the ability to *earn* a credit, not to detect fraud: the tube and the command record tie at 100% here and differ entirely in cost and in what they can prove about a doctored log. And photograph-plus-self-report scores 0% not because it is worthless but because it produces no observation series to reconcile against; every confirmation under it must come from a field visit. Radar therefore keeps the role VM0051 Appendix 4 assigns it: a per-plot cross-check that lowers the frequency of field verification [22].

S10. Harness defects and design details

S10.1 Per-event, not per-hour, recording behaviour

Rounding and delay are applied per *event*, not per hour: applying a 5 mm rounding to a 40 mm event spread over eight hours rounds each 5 mm fragment to zero and silently deletes the record, an artefact that produced empty logs in an early version of this harness.

S10.2 Two harness defects that had to be fixed

Two harness defects had to be fixed before the reported numbers meant anything, and both are recorded in the source because they are easy to reintroduce. First, a fraud generator that finds no improving move returns the *original* honest log, so an unconditional count of "attacks caught" silently includes cases where no attack occurred; every adversarial row therefore carries a `changed` flag and all attack statistics are conditioned on it. Second, comparisons must use the baseline of the variant itself: the paper-condition honest log is generated by a separate call, and scoring it against the field-condition record of a different soil realisation falsely "caught" 56 of 60 honest logs.

S10.3 Volume is a diagnostic, not a target

A configuration that maximises claimed volume is not automatically profitable: an early adversary that inflated claims by 344 mm drove n_{dry} to zero and would have earned nothing, so volume is reported only as a diagnostic.

S10.4 The self-consistency artefact

The audited layer generated "real" trajectories with the default `SoilProfile`, noiseless ERA5 forcing, irrigation as a one-hour impulse of exactly $\{20, 40\}$ mm, a threshold of exactly 0.80, and a farmer who records the right depth at the right hour every time. Its $0/60$ false accusations and $60/60$ detections were therefore both measurements of self-consistency: the detector was grading the generator that produced the "real" trajectories it audited. In the present design the field is generated by a separate process that differs from the twin along the seven axes listed in Table S5.

S11. Proofs of Propositions 1 and 2

Proof of Proposition 1 (exact blindness of aggregate defences).

$U(L) = \sum_i u_i$ is a symmetric function of the multiset, so it is invariant under any reassignment of depths to hours; M is a property of the field, not of the log; hence $G = M - U$ is unchanged. Likewise the infiltration bound (Eq. 13 of the main text) is evaluated per event and depends only on (u_i, τ_i), never on t_i or on $\hat{w}$, so the count of violations is a function of the multiset alone.

Proof of Proposition 2 (credit gain and evasion are the same action).

(i) is definitional: the set $\mathcal{A}$ requires no claim inside the interval, so removing one extends or creates a qualifying phase. For (ii), delaying a claim to $t + \delta$ means the bucket continues to lose water for δ further hours with no inflow, so $\hat{w}^-_{t+\delta} \le \hat{w}^-_t$ and $c_{t+\delta} \ge c_t$; the bound on the deficit growth follows because the loss per hour is at least the mean $\overline{\Delta}$ until field capacity is reached, after which c saturates at $FC - \hat{w}$. Since the storage-excess filter fires only when $u_i > c_{t_i} + \varepsilon$ and u_i is unchanged, a larger c_{t_i} can only reduce the violation set.

S12. Proof of Proposition 4 (completeness of the anchor)

The first clause is the definition of the anchor verdict: any disagreement of the hour sets or of a depth at a shared hour triggers forge. For the second clause, total volume, individual depths and timestamps are all functions of the hour set and the per-hour depths, so agreement on those two fixes all three, and no forgery of the three quantities has occurred.

S13. The wilting-point filter: full derivation

The layer included a third physical filter, flagging depletion below the wilting point as physically impossible. It was doubly inoperative. Its threshold was set at 1.02 while the reported indicator $\hat{d}_t = \min(1, \max(0, \cdot))$ is clipped to $[0, 1]$, so the condition was unsatisfiable by construction: 0 firings in 281 logs. We repaired it by evaluating the unclipped depletion, and then measured whether the repaired filter can activate at all. Across three stations, two climate years, five textures and five initial states — 150 configurations — with an empty log, i.e. the driest possible forcing, the unclipped depletion peaks at 0.9987 (Table S3). A dry-season delta rice field does not reach the wilting point. The filter is therefore a *safety* device that guarantees no absurd state is accepted; it is not evidence of detection, and the layer must be described as having two operative physical filters and one inert one.

S14. The two corrections to the admissible-depth bound

First, Eq. (16) of the main text uses max, not a difference: an initially dry field ($w_0 = 0.90$) admits *larger* claims than a wet one, and the original expression $(d^* - w_0)\,\text{span} + \varepsilon$ yields 43.5 mm for span = 70 mm and $w_0 = 0.25$, which corresponds to no physical state at all, since the operative deficit at that hour is either 0.25 span (initially) or d^*span (once the field has dried). Second, the bound is *per event*, not per season: the bucket drains continuously through deep percolation and evapotranspiration, so a season-long claim total is not bounded by $u_{\max}$ and an adversary issuing many sub-threshold claims can inflate the total without bound.

S15. Why the paired baseline is the honest one

Forged variants are only generated where a trajectory carries enough commands to manipulate (three or more), and those trajectories are already drier than average: their honest baseline carries 4.87 creditable dry phases against 3.23 for all honest logs. An unpaired comparison against the broader baseline therefore attributes the pre-existing dryness to the attack and overstates the inflation roughly twofold. Every figure in Table 4 of the main text is computed on the paired subset (same station, year, seed and soil), conditioned on the `changed` flag.

S16. The nine log types

Three are naive forgeries: added events on rainy days, one third of entries inflated to the whole storage span, and a 96 h window of claims deleted. Four are adversarial and, unlike the naive ones, are constructed by an optimiser rather than a fixed recipe: consolidation onto a 24 h and a 96 h grid; a rational adversary that selects whichever consolidation window maximises n_{dry} subject to preserving the total and respecting the infiltration bound; and the time-shifting hill-climb, which searches integer offsets in $[-36, 36]$ and accepts a move only if the total is preserved bit-for-bit, every depth stays below the infiltration bound, and n_{dry} strictly increases. The remaining two honest classes (paper-condition and field-condition) are the baselines.

S17. The cost of robustness, and agreement across uncertainty sets

S17.1 The measured detection loss

Replacing the point model by the intersection over Θ is not free, and the loss must be reported. On the nine omission-fraud logs that were actually altered, the point model catches $9/9$ while the robust rule catches $4/9$ (44%); under the full Θ the figures are $4/4$ and $2/4$ (50%). The same trade-off runs in the same direction as the proportional threshold, and for the same reason: a defence that refuses to accuse on the strength of one model configuration also refuses some true positives. Naive forgeries remain at 100% under both. We regard this as the correct side of the trade, since a false accusation falls on a farmer with no means of appeal while a missed forgery falls on a verification body that samples anyway, but it is a loss of detection power and not a Pareto improvement.

S17.2 Reduced and full uncertainty sets agree

Every conclusion holds under both the reduced Θ (24 replays per log, 281 logs, 10 seeds) and the full Θ (150 replays, 85 logs, 3 seeds): false accusation $0/60$ and $0/18$, anchor completeness $116/116$ and $37/37$, robust-tier time-shift escape $8/9$ and $3/3$, paired credit inflation $+33\%$ ($t = 5.29$) and $+38\%$ ($t = 3.46$). The reduced set was used for the large runs because it contains the extremes of every axis, which is where an innocent world is most likely to exist; the full set confirms that no interior configuration changes a verdict.

S18. Why the anchor does not re-introduce false accusation (full text)

Proposition 4 of the main text is strong enough to look dangerous: any discrepancy is treated as forgery, including the forgetting and rounding that the main text showed to be ubiquitous. The resolution is that the anchor does not compare a log against a *model*; it compares a transcript against the *source*. An honest farmer who forgets an entry is not accused of fraud, because the system of record is the gateway, not the farmer's memory: the claim is simply completed from $\mathcal{G}$. In

operational terms the anchor turns the log from *evidence the farmer must get right* into *a view of a record the device already holds*. False accusation then requires the gateway to disagree with itself, which is a device-integrity question rather than a behavioural one, and is why the guarantee holds at 0/60 while the behavioural tiers hold at 26.7% and 5.0%.

S19. What the attack steals

The attack does not steal water. Total pumped volume is unchanged, so the farmer pays the same and the atmosphere sees the same methane. What is stolen is *history*: the credit is issued for a dry-phase pattern that did not occur. Under VM0051 the crediting unit is precisely that pattern, so the loss is real even though no physical resource moved.

S20. Experimental design: full detail

S20.1 Removing the circularity, axis by axis

The field is generated by a separate process that differs from the twin along seven axes, each with a published source (Table S5): soil texture drawn from the five FAO-56 Table 2 classes present in the delta; K_c staged per Table 12 rather than fixed at 1.05; irrigation efficiency $\eta \in [0.60, 0.85]$, so the farmer reports water *pumped* while only ηu reaches the root zone; field rainfall drawn as 0.7–1.4 times the grid cell plus Gaussian noise; bund seepage 0–0.02 mm h^{-1}; recording behaviour with 5% forgetting, rounding to 5 mm, a 35% chance of a one-day delay and 15% depth error within $\pm$35%; and an initial depletion w_0 drawn from $[0, 0.45]$ rather than assumed to be field capacity. Rounding and delay are applied per *event*, not per hour (S10.1).

S20.2 What survives of the original claim

The premise that motivated this work was attractive: an irrigation command is already a digital event, so MRV evidence should be nearly free. That premise survives, but not in the form originally claimed. It survives because the controller that issues the command can also sign it, and the signed record is what makes the evidence admissible — not because a hydrological model can detect a lie. Where the original draft claimed that physics attests the log, the correct statement is that physics *constrains* the log: it rules out depths no soil can hold and states no field can reach, which is worth having, and it is silent about the one transformation that pays. Keeping ε secret would not close the hole either: the attack does not need to know the threshold, and a verifier must be able to reproduce a decision in any case (S9.1).

S21. Limitations (full text)

1. *Simulated farmers.* Forgeries are injected into simulated reference trajectories. A field pilot is needed to measure behavioural response, in particular whether farmers shift their *actual*

pumping once the anchor is known to exist — an attack our model cannot represent, since it changes M rather than L.

2. *The anchor assumes an honest gateway.* Proposition 4 of the main text is completeness against log forgery, not against a compromised controller. Device integrity, key custody and replay protection are assumed from the companion work and are not re-established here.

3. *Not applicable to manual irrigation.* Where a sluice is opened by hand there is no command record, so the anchor is unavailable. This excludes a substantial share of delta farms and bounds the addressable population of the scheme.

4. *The attack is demonstrated in the water-scarce year only.* In the water-surplus year (2025) farmers need so little irrigation that most simulated logs fall below the three-command minimum at which a timing attack is even constructible: of 37 genuinely altered logs under the full Θ, all 37 come from 2024. The escarpment between the two years is itself a finding — MRV fraud risk is concentrated where water is scarce — but it also means the attack results are not yet evidence about a wet season. Under the reduced Θ twelve altered logs do come from 2025, and they behave the same way, which is suggestive rather than conclusive.

5. Θ *is a judgement.* Five textures, two K_c regimes, three efficiencies and five initial states bracket the delta but do not exhaust it. A field outside Θ loses the guarantee of Proposition 3 of the main text, which is exactly the situation the proportional threshold and the query verdict exist to absorb.

6. *Device prices are planning estimates, not quotations.* The tier ordering and the break-even inequality are unaffected; the absolute margin is.

7. *No emission quantification.* The layer produces activity data. Coupling it to a methane model and locally calibrated emission factors is separate work, as is the cost table's assumption that credits scale with n_{dry}.

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